LEARNING MODULE

16

Joint Distributions and Dependence

How several random returns move together: joint and conditional laws, covariance, the multivariate normal, and what a normal model says about correlation in a crash.

VOLUME III · Probability

BUILDS ON LM1 · LM3 · LM5 · LM7 · LM8 · LM9 · LM10 · LM11 · LM12 · LM13 · LM14 · LM15

LEADS TO LM17 · LM18 · LM19 · LM20 · LM21 · LM22 · LM23 · LM24 · LM25 · LM26 · LM27 · LM28 · LM30 · LM31 · LM32 · LM33 · LM35

TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	**1** compute joint, marginal and conditional distributions from counts or a joint density, and decide whether random variables are independent
☐	**2** define population covariance and correlation, prove $\lvert\rho\rvert \le 1$, and show that uncorrelated variables can be dependent
☐	**3** compute the mean and variance of sums and portfolios, and explain why every covariance matrix is positive semidefinite
☐	**4** apply conditional expectation, the tower property and the law of total variance
☐	**5** describe the multivariate normal through linear maps, colouring, marginals and conditional distributions
☐	**6** define the chi-square law, and interpret probability contours and the squared Mahalanobis distance against it
☐	**7** compare a crash correlation with the bivariate normal benchmark under the same conditioning event
☐	**8** compute joint frequencies, conditional moments, Mahalanobis distances and normal benchmarks in Python with the right divisor, guard and tail function

LEARNING MODULE OVERVIEW

- A joint distribution holds the dependence that two marginal distributions miss. In U28, FinancialInsurance fell 1% or more on 40.4% of the days on which Semiconductor did, against 12.28% of all days.
- Uncorrelated is not independent. The TAIEX price index's daily returns correlate 0.040 with the next day's, but their sizes correlate 0.183, and after a move of 2% or more the next day's size averaged 1.28% against 0.75%.
- Covariance, correlation and covariance matrices are expectations. Under the empirical distribution they equal the sample statistics of Volumes I and II exactly, and every covariance matrix is positive semidefinite.
- The variance of a sum adds covariances, (16.3). A model year of 245 uncorrelated days with the TAIEX price index's daily log-return moments has mean 13.36% and standard deviation 17.83%.
- The tower property and the law of total variance split FinancialInsurance's daily variance of 1.482 ($\%^2$) into 1.140 within Semiconductor's three bins and 0.342 between them: 23.1%.
- A multivariate normal vector is defined by its normal combinations. Colouring, whitening and the normal pair's conditional law follow, and the squared Mahalanobis distance has a chi-square law.
- For the pair, a normal model with the sample's covariance matrix expects 48.68 days, with a standard deviation of 6.94, outside the whitened circle of radius 3; the data hold 177.
- A normal pair with a constant correlation of 0.5749 gives 0.860 on days of 2-standard-deviation moves either way and 0.156 on joint falls of 2. The pair shows 0.788 and 0.669: only the one-sided events exceed the benchmark.

1 INTRODUCTION

Volumes I and II measured co-movement in samples: LM7's covariance, LM8's correlation, LM10's covariance matrix and LM12's whitened returns. This module builds the probability model behind them, the joint distribution of several random returns. Under LM13's empirical distribution, each population formula gives back its sample formula exactly.

MOTIVATING CASE **Do correlations rise in a crash?**

Semiconductor and FinancialInsurance are two of U28's sector indices. Over 4,382 daily returns, from 2 December 2008 to 8 October 2026, their correlation is 0.5749. Over 7–9 April 2025, the three sessions after the Qingming holiday, LM10 found 0.851. On 7 April both fell nearly 10%, the pair's most extreme day of the sample.

LM12 whitened the pair. In those coordinates 177 days lie outside the circle of radius 3. If the two returns were jointly normal with the sample's covariance matrix, about 49 would lie there.

So did the correlation rise in the crash, or would any sell-off look like this? What should a crash correlation be compared with, and how many extreme days does a normal model allow?

Exhibit 1: The pair in the case

Quantity	Value
Daily returns, 2 December 2008 to 8 October 2026	4,382
Correlation $\hat{\rho}$, all days	0.5749
Correlation, 7–9 April 2025 (LM10)	0.851
Returns on 7 April 2025: Semiconductor, FinancialInsurance	−9.95%, −9.89%
The pair's d_t^2 on 7 April 2025 (rank among the days)	72.05 (1st)
Days with $d_t^2 > 9$, outside the whitened circle of radius 3	177
Normal model: probability of $d_t^2 > 9$	1.11%
Normal model: expected days; standard deviation of the count	48.7; 6.94

Note: daily simple price returns (no dividends), divisor T; d_t^2 is the squared Mahalanobis distance of Section 6. LM10's 0.851 uses its printed returns (unrounded: 0.850). Source: FinMind `TaiwanStockPrice`, retrieved 11 October 2026, with 524 closes on 38 dates of U28 repaired from the 13:30:00 values of `TaiwanStockEvery5SecondsIndex`, none on these dates or the dates before them. Computations: `data/lm16_numbers.py` and `LM16_lab.ipynb`.

Exhibit 1 collects the case's numbers. Sections 2–5 build the tools, Section 6 the multivariate normal and the case's count, Section 7 the crash correlation, and Section 8 the Python.

NOTE **Notation for this module**

Data: U28's $T = 4{,}382$ daily price returns, $A = 245.48$ (LM10). The pair is Semiconductor and FinancialInsurance; S5 adds Food, Plastics and ShippingTransportation. Moments divide by T (a bin's by its days). A model's correlation is ρ, a sample's $\hat{\rho}$.

X, Y: random variables, set in each example; $p_{X,Y}$, $f_{X,Y}$, $F_{X,Y}$, $p_{Y|X}$, $f_{Y|X}$: joint and conditional pmf, density and distribution function; $h(x) := \mathbb{E}[Y \mid X = x]$; h_i: the functions of (16.1); $\mathcal{S}$: a set of values of X, $\sigma^2_{\mathcal{S}} := \mathrm{Var}(X \mid X \in \mathcal{S})$; $\mathbf{X}$, $\mathbf{R}$: random vectors, with mean $\boldsymbol{\mu}$, covariance matrix Σ (entries σ_{ij}, $\sigma_{ii} = \sigma_i^2$) and $\mathrm{Corr}(\mathbf{X}) := \mathrm{diag}(\sigma_i)^{-1}\Sigma\,\mathrm{diag}(\sigma_i)^{-1}$, while $\hat{\Sigma}$ and C stay the sample's; a_i, $\mathbf{a}$: coefficients; M, $\mathbf{b}$: a $J \times N$ matrix and a vector; L: the Cholesky factor, $\Sigma = LL^\top$, entries l_{ij} (no leverage here); β, ε: slope and residual of a normal pair; d_t^2: day t's d^2, LM12's $\|\mathbf{z}_t\|^2$; n: the number of variables in (16.1) and (16.3), and a binomial's trials; p: the binomial's success probability. Counts: T (index t), N (indices i, j; also degrees of freedom), J, n, and m (days summed). A number of standard deviations is $k \geq 0$.

NOTE **Research hygiene**

Every count and correlation here describes U28's 4,382 days, 2 December 2008 to 8 October 2026 (sectors and the TAIEX price index): **descriptive**. The normal model and Example 5's model use the same days' moments: **in-sample** benchmarks, not forecasts. The lab's comparison of the two price-limit regimes is a **stability check**; whether gaps could be chance is LM21's and LM22's question. Example 2, Example 3, the knowledge checks of Sections 3 and 4, and Problems 2–4 and 9 are **constructed**.

2 JOINT, MARGINAL AND CONDITIONAL DISTRIBUTIONS

- ☐ compute joint, marginal and conditional distributions from counts or a joint density, and decide whether random variables are independent

Start with counts. Pick one of the 4,382 days at random, each with probability 1/4,382: the empirical distribution of LM13. Let X be Semiconductor's bin that day: −1 for a return at or below −1%, 1 at or above 1%, 0 between. Let Y be FinancialInsurance's bin, defined the same way (Exhibit 2).

Exhibit 2: Days by the bins of the two sectors' returns

Semiconductor (rows), FinancialInsurance (columns)	$R \le -1\%$	between	$R \ge 1\%$	Total
$R \le -1\%$	347	478	34	859
between	171	2,100	248	2,519
$R \ge 1\%$	20	654	330	1,004
Total	538	3,232	612	4,382

Note: U28, 4,382 days. No return of either sector is exactly ±1%, so the edges decide no day. Probabilities are counts divided by 4,382. Source: as Exhibit 1.

DEFINITION **Joint, marginal and conditional pmf**

Let X and Y be discrete random variables on one sample space. Their **joint pmf** is $p_{X,Y}(x, y) := \Pr(X = x, Y = y)$. The **marginal** pmfs are its sums over the other variable, $p_X(x) = \sum_y p_{X,Y}(x, y)$ and $p_Y(y) = \sum_x p_{X,Y}(x, y)$. For $p_X(x) > 0$, the **conditional pmf** of Y given $X = x$ is

$$p_{Y|X}(y \mid x) := \frac{p_{X,Y}(x, y)}{p_X(x)}.$$

The marginal pmfs are the table's totals. The conditional pmf is (13.4) for the events $\{X = x\}$ and $\{Y = y\}$: one row divided by its total. Read backwards, $p_{X,Y}(x, y) = p_X(x)\, p_{Y|X}(y \mid x)$ is the multiplication rule (13.5).

EXAMPLE 1

Reading the joint pmf

From Exhibit 2, find the probability that both sectors fall 1% or more, the two marginal probabilities and their product. Then find FinancialInsurance's probability of such a fall given a Semiconductor fall of 1% or more, and given a rise.

SOLUTION

The joint probability is $p_{X,Y}(-1, -1) = 347/4{,}382 = 0.0792$. The marginals are $p_X(-1) = 859/4{,}382 = 0.1960$ and $p_Y(-1) = 538/4{,}382 = 0.1228$, with product 0.0241, which independence would make the joint probability (Section 3). The joint frequency is 3.29 times the product: a descriptive ratio, not evidence (LM21).

Given a Semiconductor fall, $p_{Y|X}(-1 \mid -1) = 347/859 = 0.404$, against the marginal 0.1228. Given a rise, it is $20/1{,}004 = 0.0199$. Knowing one bin changes the probabilities of the other.

For continuous returns, probability spreads over the plane with a density, as it spread along a line in LM5.

DEFINITION **Joint and conditional density**

A **joint density** of X and Y is a function $f_{X,Y} \geq 0$ with

$$\Pr(x_1 < X \leq x_2,\ y_1 < Y \leq y_2) = \int_{x_1}^{x_2}\int_{y_1}^{y_2} f_{X,Y}(x,y)\,dy\,dx$$

for all $x_1 < x_2$ and $y_1 < y_2$. The marginal density is $f_X(x) = \int f_{X,Y}(x,y)\,dy$, and for $f_X(x) > 0$ the **conditional density** is $f_{Y|X}(y \mid x) := f_{X,Y}(x,y)/f_X(x)$. The **joint distribution function** is $F_{X,Y}(x,y) := \Pr(X \leq x, Y \leq y)$.

By Fubini's theorem (LM5), a double integral equals an iterated integral in either order when the integrand is non-negative or absolutely integrable (stated without proof; Durrett 2019, Theorem 1.7.2). A marginal density integrates the other variable out.

EXAMPLE 2

A density on the unit square

Let $f_{X,Y}(x,y) = x + y$ for $0 \leq x \leq 1$ and $0 \leq y \leq 1$, and 0 elsewhere. Check that it is a density, and find f_X, $\Pr(X \leq \tfrac12, Y \leq \tfrac12)$ and $f_{Y|X}$. Are X and Y independent?

SOLUTION

The function is non-negative, and

$$\int_0^1\int_0^1 (x+y)\,dy\,dx = \int_0^1 \left(x + \tfrac12\right) dx = 1.$$

The inner integral is the marginal, $f_X(x) = x + \tfrac12$ on $[0,1]$; by symmetry $f_Y(y) = y + \tfrac12$. The corner square has probability $\int_0^{1/2} \left(\tfrac{x}{2} + \tfrac18\right) dx = \tfrac{1}{16} + \tfrac{1}{16} = \tfrac18$.

The conditional density $f_{Y|X}(y \mid x) = (x+y)/(x+\tfrac12)$ changes with x: it is $2y$ at $x = 0$ and $\tfrac23(1+y)$ at $x = 1$. Knowing X changes the law of Y, so the two are dependent (Section 3).

PITFALL **The marginals do not determine the joint law**

The totals of Exhibit 2 also fit the table of independent bins, each cell the product of its marginals. That table puts 0.0241 in the corner where the data put 0.0792 (Example 1). Risk models with the same marginals can disagree completely about joint losses.

導讀 **聯合分布：一張表看兩個資產**

只看每個資產自己的分布（distribution），看不出它們會不會一起跌。聯合分布（joint distribution）把「兩件事同時發生」的機率排成一張表，把列（row）或行（column）加總，就得到邊際分布（marginal distribution）。條件分布（conditional distribution）是把某一列除以該列的總和：半導體跌 1% 以上的 859 天裡，金融保險也跌 1% 以上的有 347 天，比例 0.404，遠高於不分情況時的 0.1228。連續型的情形把表格換成聯合機率密度函數（joint probability density function），把加總換成積分（integral）。同樣的邊際分布可以搭配完全不同的聯合分布，所以多資產的風險模型必須描述聯合分布本身。

KNOWLEDGE CHECK 1

From Exhibit 2, what is the probability that Semiconductor rose 1% or more, given that FinancialInsurance fell 1% or more?

ANSWER

Condition on the column: $20/538 = 0.0372$, dividing by FinancialInsurance's falls, not by a row total.

3 INDEPENDENCE, COVARIANCE AND CORRELATION

- [ ] compute joint, marginal and conditional distributions from counts or a joint density, and decide whether random variables are independent
- [ ] define population covariance and correlation, prove $|\rho| \le 1$, and show that uncorrelated variables can be dependent

LM13 called discrete random variables independent when their joint pmf is the product of their pmfs, (13.13). The general definition uses distribution functions, which every random variable has.

DEFINITION **Independence and the product rule**

Random variables $X_1, \dots, X_n$ are **independent** when the first line below holds for all numbers $x_1, \dots, x_n$. Then the **product rule**, the second line, holds for all functions h_i with $\mathbb{E}|h_i(X_i)| < \infty$:

$$\begin{aligned} &\Pr(X_1 \le x_1, \dots, X_n \le x_n) \\ &\quad = \prod_{i=1}^{n} F_{X_i}(x_i), \\ &\mathbb{E}\Big[\prod_{i=1}^{n} h_i(X_i)\Big] = \prod_{i=1}^{n} \mathbb{E}[h_i(X_i)]. \end{aligned} \tag{16.1}$$

Independence holds exactly when the joint pmf, or the joint density when there is one, is the product of the marginals. The proof below covers pmfs; for densities the equivalence is stated without proof (Grinstead and Snell 1997, §4.2, Definition 4.7 and Theorem 4.2).

Proof of the equivalence for pmfs. Summing the product of the pmfs over the values up to each x_i gives the first line of (16.1). Conversely, write the first line with $x < x_1$ in place of x_1, and let x rise to x_1. By continuity (LM13), the limit is the same equation with $X_1 < x_1$ in place of $X_1 \le x_1$ on both sides. Subtracting it from the first line leaves $X_1 = x_1$ on both sides. Repeating the step for $x_2, \dots, x_n$ gives (13.13). ■

Proof of the product rule for two discrete variables. The joint pmf factorizes, so the double sum splits:

$$\begin{aligned} &\mathbb{E}[h_1(X_1)h_2(X_2)] \\ &\quad = \sum_{x_1} h_1(x_1)\, p_{X_1}(x_1) \sum_{x_2} h_2(x_2)\, p_{X_2}(x_2) \\ &\quad = \mathbb{E}[h_1(X_1)]\, \mathbb{E}[h_2(X_2)]. \end{aligned}$$

The finite expectations allow the sums in any order. A joint density splits the same way, as in LM5's Gaussian integral, and n variables repeat the step. ■

DEFINITION **Covariance and correlation**

Let X and Y have finite variances. Their **covariance** is

$$\begin{aligned} \operatorname{Cov}(X, Y) &:= \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \\ &= \mathbb{E}[XY] - \mu_X \mu_Y, \end{aligned} \tag{16.2}$$

written σ_{XY}. If also $\sigma_X, \sigma_Y > 0$, their **correlation** is $\rho_{XY} := \sigma_{XY}/(\sigma_X \sigma_Y)$. It satisfies $|\rho_{XY}| \le 1$, with equality exactly when $\Pr(Y = \eta X + \zeta) = 1$ for constants $\eta \neq 0$ and ζ.

The second form follows from linearity (14.3). The bound copies LM8's proof of (8.6), with an expectation in place of the sum over dates (Deep Dive below).

DEEP DIVE **Why $|\rho_{XY}| \le 1$, and when it equals 1** optional · click to expand

Here $\delta > 0$ and W is a random variable. For every real η,

$$\begin{aligned} 0 &\le \mathbb{E}\big[(Y - \mu_Y - \eta(X - \mu_X))^2\big] \\ &= \sigma_Y^2 - 2\eta\,\sigma_{XY} + \eta^2\sigma_X^2. \end{aligned}$$

The choice $\eta = \sigma_{XY}/\sigma_X^2$, LM8's closest multiple, gives $0 \le \sigma_Y^2 - \sigma_{XY}^2/\sigma_X^2$, so $|\rho_{XY}| \le 1$.

Equality holds exactly when this expectation is 0. A variable $W \ge 0$ with $\mathbb{E}W = 0$ is 0 with probability one: Markov's inequality (14.7) gives $\Pr(W \ge \delta) \le \mathbb{E}W/\delta = 0$, and these events increase to $\{W > 0\}$ as δ falls (LM13). So $Y = \eta X + \zeta$ with probability one, where $\zeta = \mu_Y - \eta\mu_X$ and $\eta \ne 0$ since $\sigma_Y > 0$. Conversely, $Y = \eta X + \zeta$ gives $\rho_{XY} = \eta\sigma_X^2/(\sigma_X|\eta|\sigma_X) = \pm 1$.

Independence implies zero covariance: (16.1) with h_1 and h_2 the identity gives $\mathbb{E}[XY] = \mu_X\mu_Y$. Variables with zero covariance are **uncorrelated**. The converse fails.

EXAMPLE 3

Uncorrelated but dependent

Let X take the values −1, 0 and 1 with probability 1/3 each, and let $Y = X^2$. Show that $\mathrm{Cov}(X, Y) = 0$ although Y is a function of X.

SOLUTION

By (16.2), $\mathrm{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}X\,\mathbb{E}[X^2] = 0 - 0 \times \frac{2}{3} = 0$, since $X^3 = X$ has mean 0. Yet $\Pr(Y = 0 \mid X = 0) = 1$ while $\Pr(Y = 0) = \frac{1}{3}$, so the joint pmf is not the product of the marginals. Covariance measures only the linear part of a dependence, and this symmetric dependence has none.

Market data show the same pattern. Let R_t be the TAIEX price index's return (no dividends) on day t of the U28 dates.

EXAMPLE 4

Today's move and tomorrow's

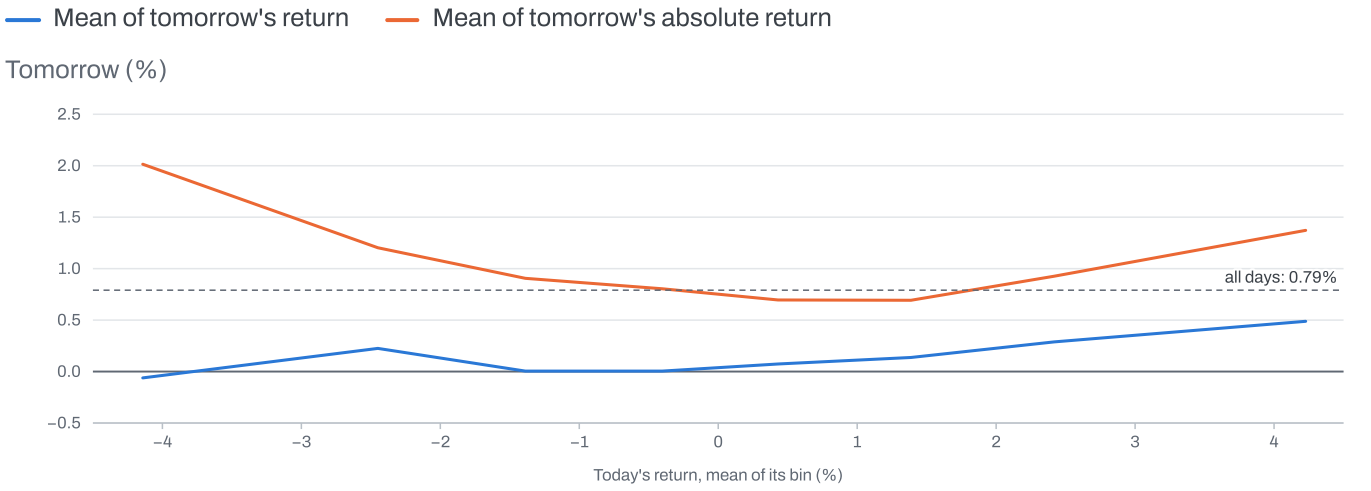
For the 4,381 pairs (R_t, R_{t+1}), compute $\hat\rho(R_t, R_{t+1})$ and $\hat\rho(|R_t|, |R_{t+1}|)$, and the mean of $|R_{t+1}|$ after the days with $|R_t| \ge 2\%$ and after the others.

SOLUTION

The returns correlate 0.040, close to 0, while their sizes correlate 0.183. After the 306 days with $|R_t| \geq 2\%$, the next day's size averaged 1.28%, against 0.75% after the others and 0.79% overall (Exhibit 3).

Under independence, both population correlations would be zero and the two conditional means equal. Large moves cluster in time, of either sign; LM26 models how tomorrow's spread depends on today's move.

Exhibit 3: The TAIEX price index: tomorrow's mean and size by today's return



Note: TAIEX price index (no dividends), U28 dates, 4,381 pairs, grouped by R_t in the bins $(-\infty, -3\%]$, $(-3\%, -2\%]$, ..., $(3\%, \infty)$ and plotted at each bin's mean R_t. The end bins hold 60 and 39 days. Dashed: the mean of $|R_{t+1}|$, 0.79%. Source: FinMind `TaiwanStockPrice`, retrieved 11 October 2026; `data/lm16_charts.py`.

Under the empirical distribution an expectation is an average with divisor T, so (16.2) is LM7's average product and ρ_{XY} is LM8's (8.9). The sample's 0.5749 is exactly the empirical model's correlation; as an estimate of another model's correlation, its error is LM20's subject. Orthogonal centred return vectors (LM8) are uncorrelated under this model, not independent.

PITFALL **A correlation near zero does not prevent a joint loss**

In 2022 0050 and 00679B both fell about 21–22% while their daily correlation was −0.06 (LM7). Correlation concerns deviations from each asset's own mean; when both means turn negative, both assets lose, whatever the correlation.

導讀 **不相關不等於獨立**

兩個隨機變數（random variable）獨立（independent），是指聯合分布（joint distribution）等於邊際分布（marginal distribution）的乘積；這時乘積的期望值（expectation）等於期望值的乘積，所以共變異數（covariance）為 0。反過來不成立：X 取 −1、0、1，$Y = X^2$，兩者互不相關（uncorrelated），Y 卻完全由 X 決定。相關係數（correlation coefficient）只量線性的部分。加權股價指數今天與明天報酬的相關係數只有 0.040，報酬大小的相關係數卻是 0.183：大波動之後常跟著大波動，方向卻難說。樣本相關係數就是經驗分布（empirical distribution）這個母體（population）的相關係數，不是另一套定義。

KNOWLEDGE CHECK 2

For constants $\eta \neq 0$ and ζ, how do $\operatorname{Cov}(X, \eta Y + \zeta)$ and the correlation of X with $\eta Y + \zeta$ compare with $\operatorname{Cov}(X, Y)$ and ρ_{XY}?

ANSWER

By (16.2), $\operatorname{Cov}(X, \eta Y + \zeta) = \mathbb{E}[(X - \mu_X)\,\eta(Y - \mu_Y)] = \eta \operatorname{Cov}(X, Y)$, since ζ cancels with the mean. The standard deviation becomes $|\eta|\sigma_Y$ (14.4), so the correlation is ρ_{XY} for $\eta > 0$ and $-\rho_{XY}$ for $\eta < 0$.

4 SUMS, PORTFOLIOS AND COVARIANCE MATRICES

- ☐ compute the mean and variance of sums and portfolios, and explain why every covariance matrix is positive semidefinite

LM7 expanded the variance of a two-asset portfolio's sample returns. The same expansion works for random variables.

THEOREM **Variance of a sum**

Let $X_1, \dots, X_n$ have $\mathbb{E}X_i^2 < \infty$ and covariances $\sigma_{ij} := \operatorname{Cov}(X_i, X_j)$, so that $\sigma_{ii} = \operatorname{Var} X_i$, and let $a_1, \dots, a_n$ be constants. Then

$$\operatorname{Var}\left(\sum_i a_i X_i\right) = \sum_i \sum_j a_i a_j \sigma_{ij}. \tag{16.3}$$

If the X_i are uncorrelated, in particular if they are independent, then $\operatorname{Var}(\sum_i X_i) = \sum_i \operatorname{Var} X_i$.

Proof. By linearity (14.3), the sum deviates from its mean by $\sum_i a_i(X_i - \mu_{X_i})$. Square it, and take expectations term by term. For uncorrelated variables only the terms with $i = j$ remain. ■

With two assets, (16.3) gives LM7's identities in population form: the portfolio variance $\sigma_p^2 = w_1^2\sigma_1^2 + 2w_1w_2\sigma_{12} + w_2^2\sigma_2^2$ and $\operatorname{Var}(R_1 - R_2) = \sigma_1^2 + \sigma_2^2 - 2\sigma_{12}$. Then (16.2) and linearity (14.3) give $\operatorname{Cov}(R_1, R_p) = w_1\sigma_1^2 + w_2\sigma_{12}$. The theorem's last sentence, $\operatorname{Var}(\sum_i X_i) = \sum_i \operatorname{Var} X_i$, needs only zero covariances, not independence, as LM3's sum of squared increments did.

EXAMPLE 5

A two-asset model

Take a model whose annualized covariance matrix for Semiconductor and FinancialInsurance is LM10's: $\sigma_1^2 = 570.34$, $\sigma_{12} = 261.83$ and $\sigma_2^2 = 363.73$, in %². Find the variance and volatility of the 50/50 portfolio, $\operatorname{Cov}(R_1, R_p)$ and the volatility of $R_1 - R_2$.

SOLUTION

By (16.3),

$$\begin{aligned}\sigma_p^2 &= 0.25(570.34) \\ &\quad + 2(0.25)(261.83) \\ &\quad + 0.25(363.73) = 364.43,\end{aligned}$$

so $\sigma_p = 19.09\%$. Next, by (16.2) and linearity, $\operatorname{Cov}(R_1, R_p) = 0.5(570.34) + 0.5(261.83) = 416.085$, exactly. Finally, $\operatorname{Var}(R_1 - R_2) = 570.34 + 363.73 - 2(261.83) = 410.41$, a volatility of 20.26%, as LM10 found for $(1, -1, 0)^\top$. Under the empirical distribution each value is also the sample value.

Sums over days use log returns, which add (LM1, (1.12)). Model the m daily log returns of a year as uncorrelated, each with mean g and standard deviation σ_r. Their sum has mean mg by (14.3) and variance $m\sigma_r^2$ by (16.3): the assumption behind LM5's annualization.

For the TAIEX price index on U28 dates, take $g = \bar{r} = 0.05454\%$ and $\sigma_r = s_r = 1.1392\%$ (divisor T), and $m = 245$, the rounded A. The model year's log return has mean 245 × 0.05454% = 13.36% and standard deviation $\sqrt{245} \times 1.1392\% = 17.83\%$ (a model fitted to the same days). If the days are iid, not only uncorrelated, a sum of many of them is nearly normal: the central limit theorem of LM17.

The same rule counts events. An indicator $\mathbf{1}_{\mathcal{A}}$ with $\Pr(\mathcal{A}) = p$ has mean p by (14.1) and, being its own square, variance $p - p^2$ by (14.4). A count of n independent such indicators has mean np and variance $np(1-p)$: LM15's binomial moments (15.2) in one line.

DEFINITION **Mean vector and covariance matrix**

Let $\mathbf{X} = (X_1, \dots, X_N)^\top$ be a random vector with $\mathbb{E}\|\mathbf{X}\|^2 < \infty$ and mean $\boldsymbol{\mu} := \mathbb{E}\mathbf{X}$. Its **covariance matrix** Σ has entries $\sigma_{ij} = \operatorname{Cov}(X_i, X_j)$. For a $J \times N$ matrix M and vectors $\mathbf{b}$ and $\mathbf{a}$ of lengths J and N,

$$\begin{aligned} \Sigma &:= \mathbb{E}\big[(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top\big], \\ \operatorname{Cov}(M\mathbf{X}+\mathbf{b}) &= M\Sigma M^\top, \\ \mathbf{a}^\top\Sigma\mathbf{a} &= \operatorname{Var}(\mathbf{a}^\top\mathbf{X}) \geq 0, \end{aligned} \tag{16.4}$$

where Cov of a vector means its covariance matrix. So Σ is positive semidefinite.

Proof. The vector $M\mathbf{X}+\mathbf{b}$ deviates from its mean $M\boldsymbol{\mu}+\mathbf{b}$ by $M(\mathbf{X}-\boldsymbol{\mu})$. By linearity, entry by entry, its covariance matrix is $\mathbb{E}[M(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top M^\top] = M\Sigma M^\top$. With $M = \mathbf{a}^\top$, a single row, this is a variance, never negative. ■

The correlation matrix $\operatorname{Corr}(\mathbf{X})$ is the covariance matrix of the entries X_i/σ_i, so it is positive semidefinite too. A portfolio return $R_p = \mathbf{w}^\top\mathbf{R}$ has variance $\sigma_p^2 = \mathbf{w}^\top\Sigma\mathbf{w}$. Under the empirical distribution, (16.4) gives LM10's (10.2), (10.3) and (10.5). Splitting σ_p^2 among assets is the work of LM27 and LM31; how far $\hat{\Sigma}$ lies from Σ is LM30's.

PITFALL **Standard deviations do not add**

Only variances and covariances combine by (16.3). The 20.26% of Example 5 is neither 23.88 − 19.07 = 4.81 points nor $\sqrt{(570.34 + 363.73)} = 30.56\%$. The first holds only when $\rho = 1$, the second only when $\rho = 0$.

KNOWLEDGE CHECK 3

Let X and Y be independent with variances 4 and 9. Find $\operatorname{Var}(X-Y)$ and the correlation of X with $X+Y$.

ANSWER

Independent variables are uncorrelated, so (16.3) gives $\operatorname{Var}(X-Y) = 4+9 = 13$: a difference adds variances too. Next, $\operatorname{Cov}(X, X+Y) = \operatorname{Var} X = 4$ and $\operatorname{Var}(X+Y) = 13$, so the correlation is $4/(2\sqrt{13}) = 0.555$.

5 CONDITIONAL EXPECTATION

- ☐ apply conditional expectation, the tower property and the law of total variance

The conditional density of Example 2 changes with x, so the mean of Y given $X = x$ changes too.

DEFINITION **Conditional expectation**

Let $\mathbb{E}|Y| < \infty$. For $p_X(x) > 0$, the **conditional expectation** of Y given $X = x$ is $h(x) = \mathbb{E}[Y \mid X = x] := \sum_y y\, p_{Y|X}(y \mid x)$; with densities, it is $\int y\, f_{Y|X}(y \mid x)\, dy$. The random variable $\mathbb{E}[Y \mid X] := h(X)$ is the conditional expectation of Y given X. The **conditional variance** $\operatorname{Var}(Y \mid X = x)$ is $\mathbb{E}[(Y - h(x))^2 \mid X = x]$, and $\operatorname{Var}(Y \mid X)$ is that function evaluated at X.

For Example 2,

$$\begin{aligned}\mathbb{E}[Y \mid X = x] &= \int_0^1 y\, \frac{x + y}{x + \tfrac{1}{2}}\, dy \\ &= \frac{x/2 + 1/3}{x + 1/2} = \frac{3x + 2}{6x + 3}.\end{aligned}$$

The conditional mean falls from 2/3 at $x = 0$ to 5/9 at $x = 1$.

On an event $\{X \in \mathcal{S}\}$ with $\Pr(X \in \mathcal{S}) > 0$, the conditional mean is the average over the event:

$$\mathbb{E}[X \mid X \in \mathcal{S}] = \frac{\mathbb{E}\big[X\, \mathbf{1}\{X \in \mathcal{S}\}\big]}{\Pr(X \in \mathcal{S})}.$$

LM5's tail mean (5.11) is this formula for a normal X and the values below a threshold.

THEOREM **Tower property and law of total variance**

If $\mathbb{E}|Y| < \infty$, then

$$\mathbb{E}\big[\mathbb{E}[Y \mid X]\big] = \mathbb{E}Y. \tag{16.5}$$

If $\mathbb{E}Y^2 < \infty$, then

$$\begin{aligned}\operatorname{Var} Y = \mathbb{E}\big[\operatorname{Var}(Y \mid X)\big] \\ + \operatorname{Var}\big(\mathbb{E}[Y \mid X]\big).\end{aligned} \tag{16.6}$$

Proof for discrete variables. The events $\{X = x\}$ partition the sample space, and total probability (13.8) gives $p_Y(y) = \sum_x p_X(x)\, p_{Y|X}(y \mid x)$. Hence

$$\begin{aligned}\mathbb{E}Y &= \sum_y y \sum_x p_X(x)\, p_{Y|X}(y \mid x) \\ &= \sum_x p_X(x)\, h(x),\end{aligned}$$

which is (16.5); the finite $\mathbb{E}|Y|$ allows the exchange of sums. For (16.6), square $Y-\mu_Y=(Y-h(X))+(h(X)-\mu_Y)$. Given $X=x$, the cross term has conditional mean $(h(x)-\mu_Y)(h(x)-h(x))=0$. Taking expectations with (16.5) leaves $\mathbb{E}[\mathrm{Var}(Y\mid X)]$ and $\mathrm{Var}(h(X))$. With densities, sums become integrals. ■

So (16.6) splits a variance into the average variance **within** the groups $\{X=x\}$ and the variance of the group means **between** them. On the empirical distribution both parts are exact sums over days.

EXAMPLE 6

FinancialInsurance's variance, split by Semiconductor's bin

Let X be Semiconductor's bin of Exhibit 2 and Y FinancialInsurance's return in %, on a day drawn from U28's empirical distribution. Use Exhibit 4 to check (16.5) and (16.6), and find the share of $\mathrm{Var}\,Y$ that the bin explains.

SOLUTION

The weights are 859, 2,519 and 1,004 days over 4,382. The tower property gives FinancialInsurance's mean:

$$\begin{aligned}&\big[859(-0.9505)+2{,}519(0.0730)\\&\quad+1{,}004(0.8458)\big]/4{,}382=0.0494\%.\end{aligned}$$

With the same weights, the squared standard deviations average 1.140 ($\%^2$), the within part. The weighted variance of the three means about 0.0494 is 0.342, the between part. Their sum, 1.482, is FinancialInsurance's variance, and the bin explains $0.342/1.482=23.1\%$ of it.

A normal model would explain $\hat\rho^2=0.3305$ with its linear conditional mean (Section 6), using Semiconductor's return itself.

Exhibit 4: FinancialInsurance by Semiconductor's bin: the law of total variance

Semiconductor's bin	Days	FinancialInsurance: mean (%)	Standard deviation (%)
$R \le -1\%$	859	−0.9505	1.4122
between	2,519	0.0730	0.8426
$R \ge 1\%$	1,004	0.8458	1.2201
All days	4,382	0.0494	1.2173

Part of $\mathrm{Var}\,Y$ (%²)	Value
Within: $\mathbb{E}[\mathrm{Var}(Y \mid X)]$	1.140
Between: $\mathrm{Var}(\mathbb{E}[Y \mid X])$	0.342
Total: $\mathrm{Var}\,Y$	1.482
Share between	23.1%

Note: U28; each bin's standard deviation divides by its days. The parts are printed to three decimals so that they add, and the printed bin values reproduce them and the weighted mean. Source: as Exhibit 1.

PITFALL **Group means are weighted by the groups' probabilities**

The plain average of the three means in Exhibit 4 is −0.0106%, the wrong sign for a sector whose mean is 0.0494%. The tower property weights each mean by $p_X(x)$, and the middle bin holds 57% of the days.

導讀 **條件期望值與全變異數法則**

條件期望值（conditional expectation）$\mathbb{E}[Y \mid X]$ 是「知道 X 之後對 Y 的平均」，它是 X 的函數，所以本身也是隨機變數（random variable）。雙重期望值定理（tower property）說，把各組的條件平均按各組的機率加權平均，就回到 $\mathbb{E}Y$。權重一定要用機率，等權平均會得到錯的答案。全變異數法則（law of total variance）把變異數（variance）拆成「組內變異數的平均」加上「組平均數（mean）之間的變異數」。以半導體的報酬分三組時，金融保險的變異數 1.482 中有 23.1% 落在組間；在經驗分布（empirical distribution）上，這兩條公式都是精確的恆等式。

KNOWLEDGE CHECK 4

LM5 found $\mathbb{E}[Z \mid Z \le -2] = -2.3732$ for a standard normal Z (its Problem 14). Write it as a conditional expectation on an event.

ANSWER

With $\mathcal{S} = (-\infty, -2]$, it is $\mathbb{E}[Z\mathbf{1}\{Z \le -2\}] / \Pr(Z \le -2)$. The numerator is $\int_{-\infty}^{-2} z\varphi(z)\,dz = -\varphi(2)$, the denominator $\Phi(-2) = 0.02275$, and the ratio -2.3732: an average over the tail.

6 THE MULTIVARIATE NORMAL DISTRIBUTION

- ☐ describe the multivariate normal through linear maps, colouring, marginals and conditional distributions
- ☐ define the chi-square law, and interpret probability contours and the squared Mahalanobis distance against it

LM12 coloured standard normal draws with a Cholesky factor to simulate returns. This section defines the law of such draws.

DEFINITION **Multivariate normal**

A random vector $\mathbf{X}$ of length N is **multivariate normal** when every combination $\mathbf{a}^\top \mathbf{X}$ is normal, a constant counting as normal with variance 0. Its law is written $\mathcal{N}_N(\boldsymbol{\mu}, \Sigma)$, with $\boldsymbol{\mu}$ and Σ as in (16.4).

Two facts are stated without proof: independent normal variables are jointly normal, and $\boldsymbol{\mu}$ and Σ fix a normal vector's law. Durrett (2019, §3.10 and Exercise 3.10.8) gives both for every positive semidefinite Σ, through characteristic functions.

THEOREM **Linear maps, colouring and the density**

Let $\mathbf{X} \sim \mathcal{N}_N(\boldsymbol{\mu}, \Sigma)$ with $\Sigma = LL^\top$, and let M be a $J \times N$ matrix and $\mathbf{b}$ a vector of length J. Then

$$\begin{aligned} M\mathbf{X} + \mathbf{b} &\sim \mathcal{N}_J(M\boldsymbol{\mu} + \mathbf{b},\ M\Sigma M^\top), \\ \boldsymbol{\mu} + L\mathbf{Z} &\sim \mathcal{N}_N(\boldsymbol{\mu}, \Sigma) \ \text{ for } \ \mathbf{Z} \sim \mathcal{N}_N(\mathbf{0}, I). \end{aligned} \tag{16.7}$$

If Σ is positive definite, $\mathbf{X}$ has the density

$$\begin{gathered} f_{\mathbf{X}}(\mathbf{x}) = (2\pi)^{-N/2} (\det \Sigma)^{-1/2} e^{-d^2/2}, \\ d^2 := (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}), \end{gathered}$$

where d^2 is the squared **Mahalanobis distance** (stated without proof; Wasserman 2004, §2.10, equation (2.10) and Theorem 2.43).

Proof of the two lines. For any $\mathbf{a}$, the variable $\mathbf{a}^\top(M\mathbf{X} + \mathbf{b}) = (M^\top \mathbf{a})^\top \mathbf{X} + \mathbf{a}^\top \mathbf{b}$ is normal, so $M\mathbf{X} + \mathbf{b}$ is multivariate normal, with the moments of (16.4). Independent standard normals form $\mathbf{Z} \sim \mathcal{N}_N(\mathbf{0}, I)$ by the first stated fact; take $M = L$ and $\mathbf{b} = \boldsymbol{\mu}$. ■

The second line is **colouring** in population form, and $L^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \mathcal{N}_N(\mathbf{0}, I)$ is **whitening**. LM12's whitened returns are only uncorrelated with variance 1, in the sample; the model's $\mathbf{Z}$ has independent normal entries. Drawing $\mathbf{Z}$ on a computer is LM18's subject.

With M a row of the identity, (16.7) makes each X_i normal; with several rows, any sub-vector is multivariate normal. For an orthogonal matrix such as LM11's Q, (16.7) gives $Q\mathbf{Z} \sim \mathcal{N}_N(\mathbf{0}, I)$: a standard normal vector looks the same in every rotated frame.

If Σ is diagonal with every $\sigma_i > 0$, $d^2 = \sum_i (x_i - \mu_i)^2/\sigma_i^2$ and the density factorizes, so uncorrelated entries of a normal vector are independent. An entry of variance 0 is constant with probability one (Section 3's Deep Dive), so it is independent of the others too.

DEEP DIVE **The density for $N = 2$** optional · click to expand

Here $\mathbf{X} = \boldsymbol{\mu} + L\mathbf{Z}$ by colouring, and $\mathbf{x} = \boldsymbol{\mu} + L\mathbf{z}$ is the image of a point $\mathbf{z}$ of the plane. The vector $\mathbf{Z}$ has the density $\varphi(z_1)\varphi(z_2) = e^{-\|\mathbf{z}\|^2/2}/(2\pi)$. The map sends a small square at $\mathbf{z}$ to a parallelogram spanned by L's columns, so areas grow by $|\det L| = l_{11}l_{22}$ (LM9, (9.2)). Both small regions have the same probability, so

$$f_{\mathbf{X}}(\mathbf{x})\, l_{11} l_{22} = \frac{e^{-\|\mathbf{z}\|^2/2}}{2\pi}.$$

Since $\det \Sigma = (l_{11}l_{22})^2$ and $\|\mathbf{z}\|^2 = d^2$ by (12.9), this is (16.7)'s density for $N = 2$.

THEOREM **The conditional law in a normal pair**

Let (X, Y) be a normal pair with $\sigma_X, \sigma_Y > 0$ and correlation ρ, and let $\beta := \sigma_{XY}/\sigma_X^2$. Then

$$\begin{aligned} Y &= \mu_Y + \beta(X - \mu_X) + \varepsilon, \\ \varepsilon &\sim \mathcal{N}\big(0, \sigma_Y^2(1 - \rho^2)\big), \end{aligned} \tag{16.8}$$

with ε independent of X. Given $X = x$, Y is normal with mean $\mu_Y + \beta(x - \mu_X)$ and variance $\sigma_Y^2(1 - \rho^2)$.

Proof. Define ε by (16.8). The pair (X, ε) is a linear map of (X, Y) plus a constant, so it is jointly normal, as in the proof of (16.7). Since $\operatorname{Cov}(X, \varepsilon) = \sigma_{XY} - \beta\sigma_X^2 = 0$, X and ε are independent. By (16.3),

$$\begin{aligned} \operatorname{Var}\varepsilon &= \sigma_Y^2 - 2\beta\sigma_{XY} + \beta^2\sigma_X^2 \\ &= \sigma_Y^2 - \frac{\sigma_{XY}^2}{\sigma_X^2} = \sigma_Y^2(1 - \rho^2). \end{aligned}$$

Fixing $X = x$ leaves the law of the independent ε unchanged. ■

The slope β is the population slope that LM12's least squares estimate; its standard error is LM23's. Since $\operatorname{Var}(\mathbb{E}[Y \mid X]) = \beta^2\sigma_X^2 = \rho^2\sigma_Y^2$, the share between in (16.6) is ρ^2.

EXAMPLE 7

Semiconductor falls 5%

Model the pair as normal with the sample's daily moments (divisor T): means 0.0947% and 0.0494%, standard deviations 1.5243% and 1.2173%, $\rho = 0.5749$. Given a Semiconductor return of −5%, find FinancialInsurance's conditional mean, standard deviation and probability of a fall of 3% or more, and compare the data.

SOLUTION

The slope is $\beta = 0.5749 \times 1.2173/1.5243 = 0.4591$. By (16.8), the conditional mean, the standard deviation (LM12's l_{22}) and the probability are

$$\begin{aligned}
&\mathbb{E}[Y \mid X = -5\%] \\
&\quad = 0.0494 + 0.4591 \times (-5 - 0.0947) \\
&\quad = -2.29\%, \\
&1.2173\sqrt{1 - 0.5749^2} = 0.996\%, \\
&\Pr(Y \le -3\% \mid X = -5\%) \\
&\quad = \Phi\big((-3 + 2.29)/0.996\big) \\
&\quad = \Phi(-0.713) = 0.238.
\end{aligned}$$

Unrounded inputs agree.

On the 17 days with Semiconductor between −5.5% and −4.5%, FinancialInsurance averaged −2.86%, with a standard deviation of 2.00% (divisor 17). Six fell 3% or more: lower and wider than the model, on few days.

DEFINITION **Chi-square law**

Let $Z_1, \dots, Z_N$ be independent standard normal variables. The law of $Z_1^2 + \cdots + Z_N^2$ is the **chi-square distribution** with N **degrees of freedom**, χ^2_N. Its mean is N, since $\mathbb{E}Z_i^2 = 1$ (LM5). Its variance is $2N$, since $\operatorname{Var}(Z_i^2) = 2$ by (14.6) and the squares are uncorrelated by (16.1), so (16.3) applies.

THEOREM **Mahalanobis distance of a normal vector**

If $\mathbf{X} \sim \mathcal{N}_N(\boldsymbol{\mu}, \Sigma)$ with Σ positive definite, the squared Mahalanobis distance of $\mathbf{X}$ has the law χ^2_N. For $N = 2$,

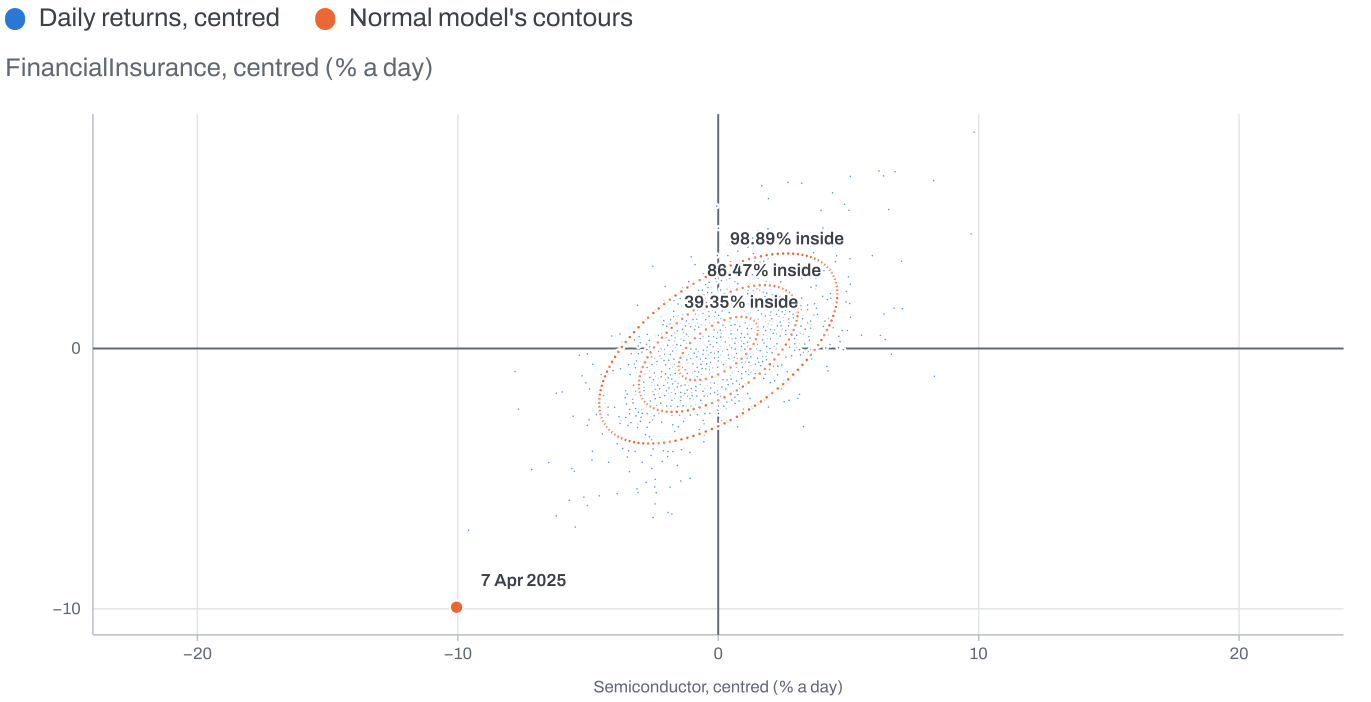
$$\Pr(d^2 > k^2) = e^{-k^2/2}. \tag{16.9}$$

Proof. Whitening gives $\mathbf{Z} = L^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \mathcal{N}_N(\mathbf{0}, I)$, and $d^2 = \|\mathbf{Z}\|^2$ because $\Sigma^{-1} = L^{-\top}L^{-1}$, as in (12.9). For $N = 2$, LM5's disc integral gives $\Pr(Z_1^2 + Z_2^2 \le k^2) = 1 - e^{-k^2/2}$. ■

For $N > 2$, SciPy's `stats.chi2` gives the tails of χ^2_N, and LM21 builds tests on it.

The density (16.7) is constant where d^2 is. With Σ's eigenpairs, (11.3) gives $\Sigma^{-1} = \sum_i \mathbf{q}_i\mathbf{q}_i^\top/\lambda_i$, so the contour $d^2 = k^2$ is an ellipse (for $N > 2$, an ellipsoid) with semi-axes $k\sqrt{\lambda_i}$ along the eigenvectors $\mathbf{q}_i$ of (11.1). It holds the probability $\Pr(\chi^2_N \le k^2)$. For the pair, $\hat{\Sigma}$ has entries 2.323354, 1.066603 and 1.481729 in %² (LM12), and (11.2) gives $\lambda_1 = 3.049$ and $\lambda_2 = 0.756$. LM11's ellipse on LM7's sample was a description; under a normal model it becomes a probability contour (Exhibit 5).

Exhibit 5: Probability contours of a normal model for the pair



Note: centred daily returns of Semiconductor (x) and FinancialInsurance (y), % a day, 4,382 days. The model's contours $d^2 = k^2$ for $k = 1$, 2, 3 hold 39.35%, 86.47% and 98.89%; the data put 56.16%, 87.95% and 95.96% of the days inside. Semi-axes $k\sqrt{\lambda_i}$, with $\sqrt{\lambda_1} = 1.746\%$ and $\sqrt{\lambda_2} = 0.869\%$; LM12's whitening exhibit drew $k = 3$ only. Source: as Exhibit 1.

EXAMPLE 8

Days outside the circle of radius 3

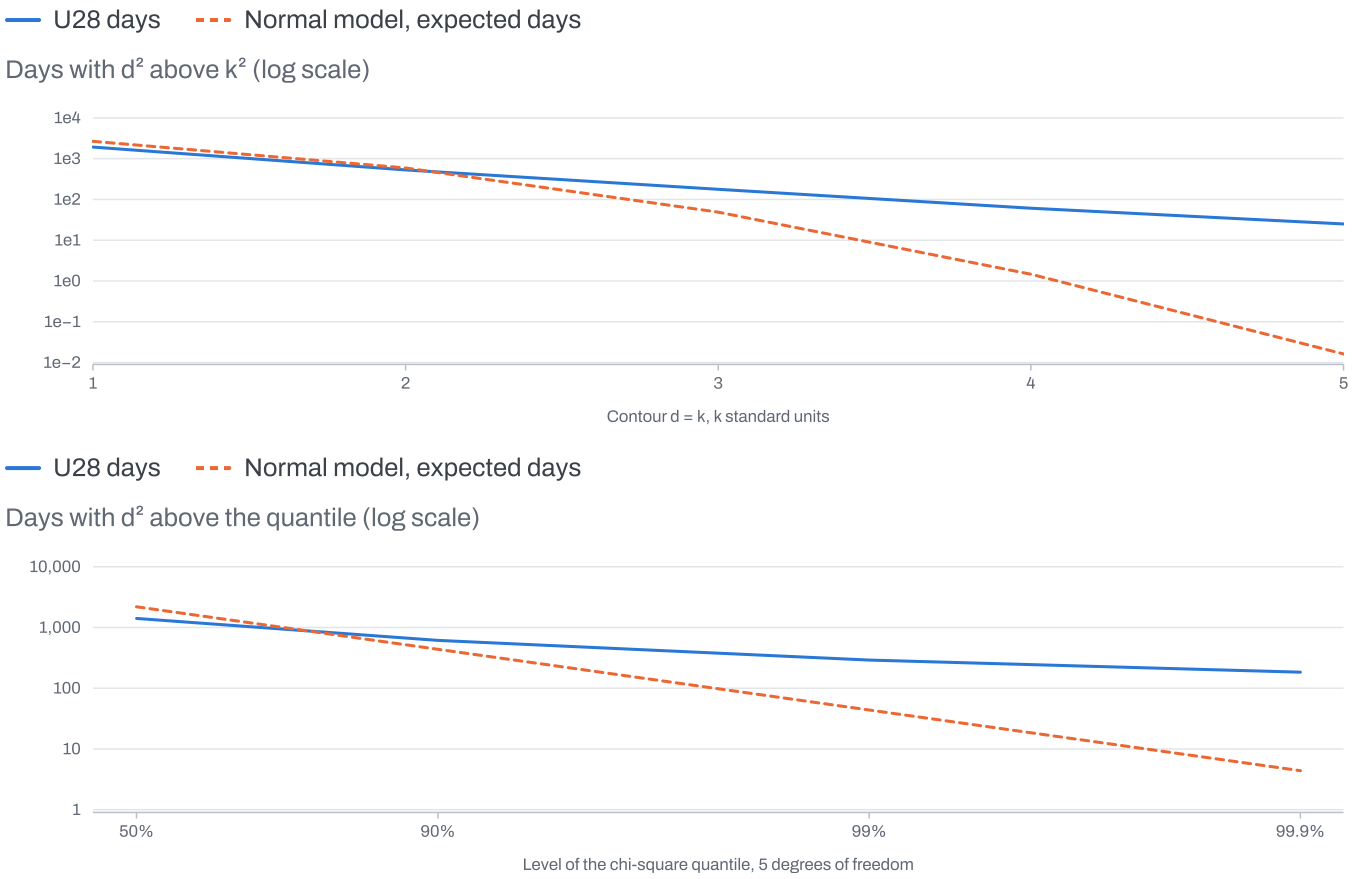
Under the normal model for the pair, find the probability that a day lies outside the whitened circle of radius 3. Among $n = 4{,}382$ independent days, find the mean and standard deviation of the count of such days.

SOLUTION

By (16.9), $p = \Pr(d^2 > 9) = e^{-4.5} = 0.011109$, about 1.11%. The count of n independent indicators is $\text{Bin}(n, p)$ ((15.1), (15.2)), with mean $np = 48.68$, variance $np(1-p) = 48.14$ and standard deviation 6.94.

The data hold 177 such days (Exhibit 6), set beside the model's mean and spread, not divided by them. Whether such a gap could be chance is LM21's question, and the model's Σ comes from the same days.

Exhibit 6: Days beyond the normal model's contours: the pair and S5



The pair: days with $d_t^2 > k^2$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
Model probability, $e^{-k^2/2}$	60.65%	13.53%	1.11%	0.0335%	0.00037%
Model, expected days	2,657.8	593.0	48.7	1.47	0.016
U28 days	1,921	528	177	61	25
S5: days above the χ^2_5 quantile		**50%**	**90%**	**99%**	**99.9%**
Quantile		4.35	9.24	15.09	20.52
Model, expected days		2,191	438.2	43.8	4.4
U28 days		1,411	613	291	184

Note: log scale. The mean of d_t^2 is exactly 2 and 5 by (12.9), so only the shape is compared. Source: as Exhibit 1.

EXAMPLE 9

How rare is 7 April 2025 under a normal model?

LM12 found $d^2 = 89.30$ for S5 on 7 April 2025, the largest of the 4,382 days. Under a normal model with S5's sample moments, how likely is a day at least that far out?

SOLUTION

By (16.9), $d^2 \sim \chi^2_5$, and `stats.chi2.sf(89.30, 5)` gives 9.4×10^{-18}: at $A = 245.48$, once in about 4×10^{14} years. 184 days lie beyond the 99.9% quantile 20.52, where the model expects 4.4 (Exhibit 6). The 10% limit capped the day (LM12), so even 89.30 understates the shock.

Exhibit 7 lists the population counterparts of earlier sample results.

Exhibit 7: Sample results and their population counterparts

Sample result (Volumes I–II)	Population counterpart (this module)
LM7's average product of deviations	$\mathrm{Cov}(X, Y)$, (16.2)
Correlation as a cosine, (8.9)	ρ_{XY}, (16.2)
Cauchy–Schwarz, (8.6)	$\lvert\rho_{XY}\rvert \le 1$, Section 3
LM7's two-asset variance	variance of a sum, (16.3)
$\hat{\Sigma}$ and C, (10.2) and (10.3)	Σ and $\mathrm{Corr}(\mathbf{X})$, (16.4)
$\mathbf{w}^\top \hat{\Sigma} \mathbf{w} \ge 0$, (10.4) and (10.5)	$\mathrm{Var}(\mathbf{w}^\top \mathbf{R}) = \mathbf{w}^\top \Sigma \mathbf{w} \ge 0$, (16.4)
Mean of $\lVert\mathbf{z}_t\rVert^2$ equal to N, (12.9)	$\mathrm{Cov}(L^{-1}(\mathbf{X} - \boldsymbol{\mu})) = I$, so $\mathbb{E}d^2 = N$, (16.4)
Colouring with L (LM12)	$\boldsymbol{\mu} + L\mathbf{Z} \sim \mathcal{N}_N(\boldsymbol{\mu}, \Sigma)$, (16.7)

Note: under the empirical distribution, expectations are averages with divisor T, and the right column gives the left exactly. The last row also needs a normal $\mathbf{Z}$.

PITFALL **Normal marginals do not make a normal vector**

A pair can have two standard normal marginals and zero covariance and still be dependent. The rule "uncorrelated implies independent" needs the pair to be jointly normal, not only each marginal.

導讀 **多變量常態分布與 Mahalanobis 距離**

多變量常態分布（multivariate normal distribution）的定義是：任何線性組合（linear combination）都是常態分布（normal distribution）。把獨立（independent）的標準常態（standard normal）變數乘上 Cholesky 因子 L 再加上 $\boldsymbol{\mu}$，就得到共變異數矩陣（covariance matrix）為 Σ 的常態向量，這是著色（colouring）；反過來乘上 L^{-1} 就是白化（whitening）。LM12 的樣本白化後只保證互不相關（uncorrelated）且變異數（variance）為 1，模型裡的 $\mathbf{Z}$ 卻是獨立的常態，兩者差很多。常態向量的 Mahalanobis 距離（Mahalanobis distance）平方服從自由度（degrees of freedom）為 N 的卡方分布（chi-square distribution），二維時落在半徑 k 之外的機率是 $e^{-k^2/2}$。半導體與金融保險這一對，模型預期約 49 天落在半徑 3 之外，實際有 177 天；這個差距是否可能出於偶然，留待 LM21。

KNOWLEDGE CHECK 5

For S5, 7 April 2025 has the sample's largest d_t^2. Among U28's 28 sectors its d_t^2 is 132.29 and ranks only 15th; the largest, 209.55, falls on 5 May 2009. Why?

ANSWER

Deviations along directions of little variance count far more. On 7 April 2025, 97.1% of the squared deviation lies along the first eigenvector of $\hat{\Sigma}$, LM11's common move, whose variance is large. On 5 May 2009 the sectors ran from −3.71% to +6.06% while the TAIEX price index rose 0.78%, a dispersion that the small eigenvalues rarely allow.

7 CORRELATION IN A CRASH

- [ ] compare a crash correlation with the bivariate normal benchmark under the same conditioning event

The case's three days gave a correlation of 0.851, against 0.5749 for the whole sample. All 28 sectors fell together on 156 days (LM11). A higher correlation on bad days looks like contagion. Yet a normal pair with one fixed ρ also shows another correlation on selected days.

THEOREM **Correlation given an event of one variable**

Let (X, Y) be a normal pair with $\sigma_X, \sigma_Y > 0$ and correlation ρ. Let $\{X \in \mathcal{S}\}$ be an event with $\Pr(X \in \mathcal{S}) > 0$ on which X has a positive conditional variance $\sigma_{\mathcal{S}}^2$. Then

$$\begin{aligned} &\operatorname{Corr}(X, Y \mid X \in \mathcal{S}) \\ &\quad = \frac{\rho}{\sqrt{\rho^2 + (1 - \rho^2)\, \sigma_X^2 / \sigma_{\mathcal{S}}^2}}. \end{aligned} \tag{16.10}$$

(Boyer, Gibson and Loretan 1999, Theorem 1.)

Proof. By (16.8), $Y = \mu_Y + \beta(X - \mu_X) + \varepsilon$ with ε independent of X, hence of the event, on which it keeps its mean 0 and variance $\sigma_Y^2(1 - \rho^2)$. Conditionally, $\operatorname{Cov}(X, Y) = \beta \sigma_{\mathcal{S}}^2$ and $\operatorname{Var} Y = \beta^2 \sigma_{\mathcal{S}}^2 + \sigma_Y^2(1 - \rho^2)$ by (16.3). Divide by the two standard deviations, put $\beta = \rho \sigma_Y / \sigma_X$, and cancel $\sigma_Y \sigma_{\mathcal{S}} / \sigma_X$. ■

The correlation rises when the event widens the spread of X and falls when it narrows it. Moves of k standard deviations either way widen it; a one-sided tail, whose values lie close together, narrows it. Forbes and Rigobon (2002) adjust crisis correlations this way before reading them as contagion.

DEEP DIVE **Conditional variances of a standard normal** optional · click to expand

Here $\ell := \varphi(k)/\Phi(-k)$ for $k \geq 0$. Integration by parts, as LM5 found $\mathbb{E}Z^2 = 1$, gives

$$\begin{aligned} \int_{-\infty}^{-k} z^2 \varphi(z)\, dz &= \big[-z\varphi(z) \big]_{-\infty}^{-k} \\ &\quad + \int_{-\infty}^{-k} \varphi(z)\, dz \\ &= k\varphi(k) + \Phi(-k). \end{aligned}$$

Dividing by $\Phi(-k)$ gives $\mathbb{E}[Z^2 \mid Z \leq -k] = 1 + k\ell$, and (5.11) gives $\mathbb{E}[Z \mid Z \leq -k] = -\ell$. Hence

$$\begin{aligned} \operatorname{Var}(Z \mid Z \leq -k) &= 1 + k\ell - \ell^2, \\ \operatorname{Var}(Z \mid |Z| \geq k) &= 1 + k\ell, \end{aligned}$$

the second because the two-sided event has mean 0 and equal tails. At $k = 2$, $\ell = 2.3732$ (LM5, Problem 14), and the variances are 0.1143 and 5.7464, which enter (16.10) with $\sigma_X^2 = 1$.

Exhibit 8 applies (16.10) to the pair with $\rho = 0.5749$ and sets the data beside it.

Exhibit 8: Correlation of the pair given Semiconductor's move

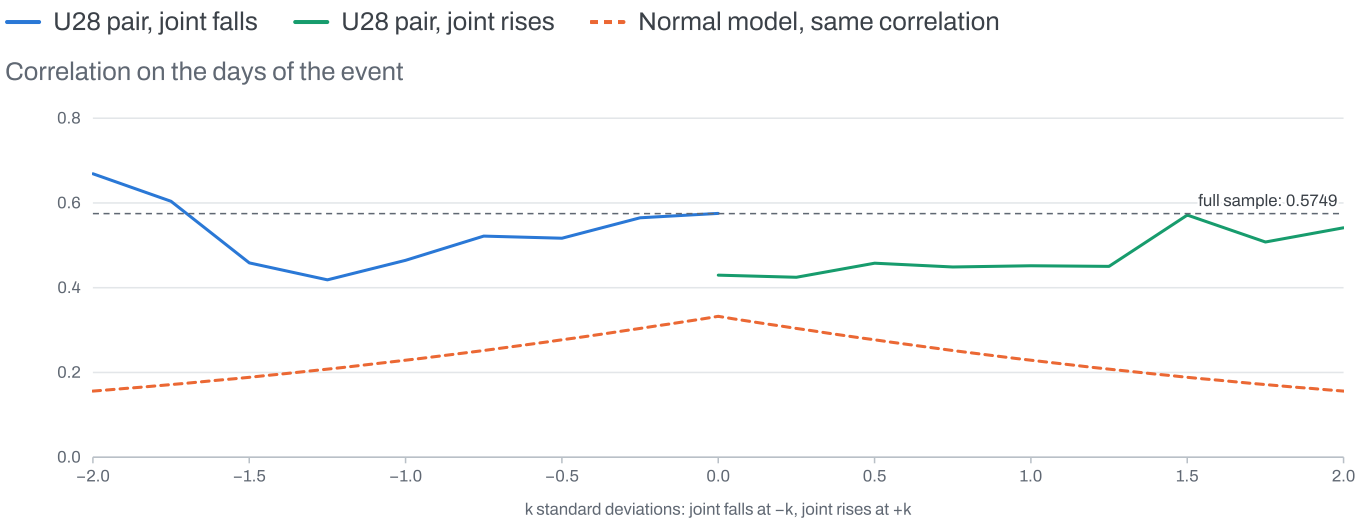
Event for Semiconductor (k standard deviations)	k	Days	Data $\hat{\rho}$	Normal model (16.10)
moves k or more either way	1	1,088	0.692	0.745
	2	223	0.788	0.860
	2.5	111	0.811	0.894
falls k or more	1	550	0.474	0.299
	2	105	0.488	0.231
	2.5	47	0.485	0.205
rises k or more	1	538	0.367	0.299
	2	118	0.429	0.231
	2.5	64	0.382	0.205

k	**1**	**2**	**2.5**
$\mathrm{Var}(Z \mid \lvert Z\rvert \geq k)$	2.5251	5.7464	8.0569
$\mathrm{Var}(Z \mid Z \leq -k)$	0.1991	0.1143	0.0890

Note: U28; Semiconductor standardized with full-sample moments (divisor T), the guard at each edge. The model uses $\rho = 0.5749$, and rises take the values of falls by symmetry. In-sample. Source: as Exhibit 1.

Those events condition on one variable. An **exceedance correlation** conditions on both: both returns lie k standard deviations or more below their means, or both above (Longin and Solnik 2001; Ang and Chen 2002). For a normal pair and $k > 0$ it has no closed form in φ and Φ; the script integrates it with `quad`, as LM5 did (Exhibit 9).

Exhibit 9: Exceedance correlations of the pair against the normal benchmark



	$k = 0$	$k = 1$	$k = 2$
Joint falls: days	1,509	228	44
Joint falls: $\hat{\rho}$	0.575	0.465	0.669
Joint rises: days	1,523	170	32
Joint rises: $\hat{\rho}$	0.430	0.452	0.541
Normal benchmark	0.332	0.229	0.156
Model: probability of a joint fall	0.3475	0.0699	0.0051
Model: expected joint falls (days)	1,522.7	306.3	22.4

Note: a point at $-k$ uses the days when both returns are at or below their means minus k standard deviations; a point at $+k$, at or above plus k. Full-sample moments, divisor T, $k = 0, 0.25, \ldots, 2$, at least 30 days at each point. Benchmark and model rows at $\rho = 0.5749$, by numerical integration, in-sample. The dashed horizontal line is $\hat{\rho} = 0.5749$. Source: as Exhibit 1.

EXAMPLE 10

Did the correlation rise in the crash?

Express Semiconductor's moves on 7–9 April 2025 in standard deviations, from LM10's returns and Example 7's moments. Then compare the pair's conditional correlations with the normal benchmark under the same events.

SOLUTION

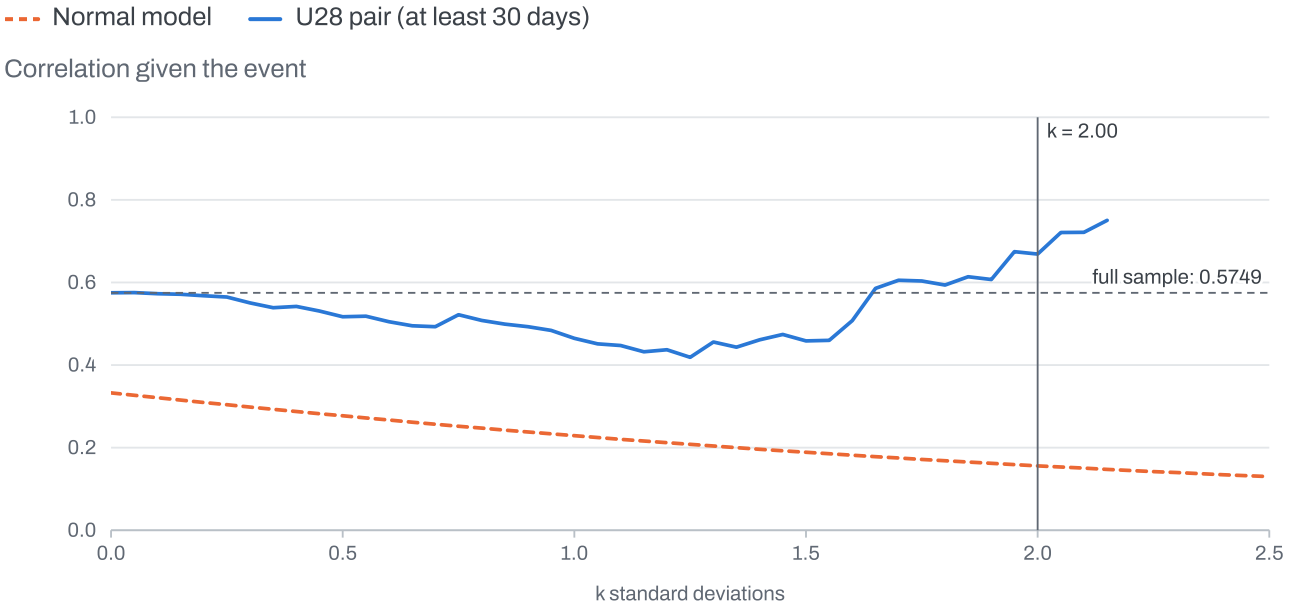
The moves are $(-9.95 - 0.0947)/1.5243 = -6.59$, −2.68 and −2.99 standard deviations (unrounded inputs give −3.00 for 9 April). Three days cannot measure a correlation, and the first, the sample's extreme, was capped by the 10% limit. Use Exhibit 8 and Exhibit 9 instead.

When Semiconductor moved 2 standard deviations either way (223 days), the data give 0.788 against the model's 0.860; at 1, 0.692 against 0.745. Larger moves alone account for these correlations above 0.5749.

One-sided events differ. Semiconductor's falls of 2 give 0.488 against 0.231 (105 days). Joint falls of 2 give 0.669 against 0.156 (44 days), and joint rises 0.541 against 0.156 (32 days). Falls exceed the model more than rises, as Longin and Solnik (2001) report, but on few days.

The widget sets the event, ρ and k. Its model values for joint events are interpolated in ρ from a grid, with errors below 10^{-5} in the correlation and 0.01 days in the count. It opens on joint falls of 2 at $\rho = 0.5749$: the model gives 0.156 with probability 0.0051 (22.4 days), the pair 0.669 on 44 days.

What a normal model says about correlation in a crash INTERACTIVE



Model correlation

0.156

Model probability · expected days in 4,382

0.0051 · 22.4

U28 pair: correlation · days

0.669 · 44

Full-sample correlation of the pair

0.5749

The model is a normal pair with the chosen correlation; at 0.5749 it is fitted to the same days as the data (in-sample). Joint events are interpolated from a precomputed grid, one-variable events use (16.10). The pair's correlation is shown where at least 30 of its 4,382 days fall in the event.

Models that keep each marginal and change only the dependence, notably in the tails, are called copulas; this book does not use them.

PITFALL

Compare a conditional correlation with the model's, not with the full sample's

The two-sided 0.788 exceeds the full-sample 0.5749, yet a normal pair with the constant correlation 0.5749 gives 0.860 for the same event. Compare a conditional correlation with the model under the same event; whether a gap could be chance is for LM21 and LM22.

導讀 **崩盤時的相關係數：和同一事件下的模型比**

只挑大波動的日子算相關係數（correlation coefficient），常態模型本身就會給出不同的數字：條件相關係數（conditional correlation）取決於條件事件（conditioning event）下 X 的變異數（variance）。雙邊的大波動讓 X 的變異數變大，相關係數跟著上升；單邊的尾部讓變異數變小，相關係數反而下降。所以崩盤時的相關係數要和「同一事件下的常態模型」比，不能和全樣本的 0.5749 比。半導體大漲或大跌 2 個標準差（standard deviation）時，資料是 0.788，模型是 0.860；兩者同時下跌 2 個標準差時，資料是 0.669，模型只有 0.156。在 2 個標準差時，超越相關係數（exceedance correlation）的下跌端（0.669）高於上漲端（0.541），方向與 Longin 和 Solnik（2001）一致，但這些天數很少。

KNOWLEDGE CHECK 6

In a normal pair with $\rho = 0.5749$, what is the correlation on the days when $|X - \mu_X| \geq \sigma_X$, given $\mathrm{Var}(Z \mid |Z| \geq 1) = 2.5251$?

ANSWER

By (16.10),

$$\frac{0.5749}{\sqrt{0.5749^2 + (1 - 0.5749^2)/2.5251}} = 0.745.$$

The data give 0.692 on 1,088 days (Exhibit 8).

8 JOINT DISTRIBUTIONS IN PYTHON

- [ ] compute joint frequencies, conditional moments, Mahalanobis distances and normal benchmarks in Python with the right divisor, guard and tail function

Counts at an edge use a guard, because index levels have two decimals and a return can equal −1% exactly (LM5). `np.cov` divides by $T-1$ unless `ddof=0` (or `bias=True`) is passed, while (10.2) divides by T. Mahalanobis distances come from a Cholesky factor and a triangular solve, never from an inverse.

PYTHON

```python
import numpy as np
from scipy import stats
from scipy.linalg import solve_triangular

# x1, x2: the pair's T daily returns (Semiconductor, FinancialInsurance)
EPS = 1e-12                                  # guard for edges such as -1%
both = (x1 <= -0.01 + EPS) & (x2 <= -0.01 + EPS)
print(both.sum())                            # 347 days, the corner cell
XP = np.column_stack([x1, x2])
S = np.cov(XP, rowvar=False, ddof=0)         # divisor T, as (10.2)
L = np.linalg.cholesky(S)                    # lower factor, S = L L'
Z = solve_triangular(L, (XP - XP.mean(axis=0)).T, lower=True)
d2 = (Z**2).sum(axis=0)                      # d_t^2 of every day
print((d2 > 9 + EPS).sum())                  # 177
print(len(d2) * stats.chi2.sf(9, 2))         # 48.68 expected
print(1 - stats.chi2.cdf(89.30, 5))          # 0.0: the tail is lost
print(stats.chi2.sf(89.30, 5))               # 9.4e-18
```

PITFALL **A tail probability from `1 - cdf` can vanish**

`stats.chi2.cdf(89.30, 5)` differs from 1 by less than 10^{-17}. Double precision has no number between $1-1.1\times10^{-16}$ and 1, so the result is stored as 1 and `1 - cdf` returns 0.0. `stats.chi2.sf` computes the upper tail directly: 9.4×10^{-18}.

IN PYTHON **Companion lab**

The Polars-first notebook `LM16_lab.ipynb` repairs the sector closes, rebuilds the module's exhibits and computational examples, and solves the computational practice problems. It ends with a stability check that compares the two price-limit regimes.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 7

Why does `np.corrcoef` need no `ddof` argument?

ANSWER

The divisor appears once in the covariance and once, through the two square roots, in the product of standard deviations, so it cancels, as in (8.9).

SUMMARY

- A conditional law divides a slice of the joint law by its marginal: FinancialInsurance fell 1% or more on 0.404 of Semiconductor's falls, against 0.1228 of all days.
- Independence makes covariance zero, (16.1), but not conversely: the TAIEX price index's returns correlate 0.040 day to day, their sizes 0.183.
- Covariance and correlation (16.2) are the population versions of LM7's and LM8's statistics.
- Variances of sums follow (16.3); covariance matrices are positive semidefinite, (16.4).
- By (16.5) and (16.6), Semiconductor's bin explains 23.1% of FinancialInsurance's variance.
- By (16.9), a normal model expects 48.68 days outside radius 3 for the pair (standard deviation 6.94); the data hold 177.
- Conditioning changes a normal pair's correlation, (16.10): 0.860 for moves of 2 either way and 0.156 for joint falls of 2, against the pair's 0.788 and 0.669.

PRACTICE PROBLEMS

1. From Exhibit 2, the probability that FinancialInsurance falls 1% or more, given that Semiconductor fell 1% or more, is *closest* to:

   A. 0.079

   B. 0.404

   C. 0.645

2. For the density $f_{X,Y}(x,y) = x + y$ of Example 2, $\operatorname{Cov}(X,Y)$ is *closest* to:

   A. −0.0069

   B. 0.3333

   C. 0

3. Let $X \sim \mathcal{N}(0,1)$, let S be $+1$ or -1 with probability 1/2 each, independent of X, and let $Y = SX$. Which statement is true?

A. The variables X and Y are independent, since they are uncorrelated and each is normal.

B. The pair (X, Y) is jointly normal with $\rho = 0$.

C. The variable Y is normal and $\mathrm{Cov}(X, Y) = 0$, yet X and Y are dependent.

4. Two assets have volatilities $\sigma_1 = 20\%$ and $\sigma_2 = 15\%$, correlation 0.5 and weights 0.6 and 0.4. Compute the volatility of $R_1 - R_2$, $\mathrm{Cov}(R_1, R_p)$ in decimal units (four decimals), σ_p and $\mathrm{Corr}(R_1, R_p)$ (three decimals). Why is the first neither 5% nor 25%?

5. In Section 4's model year ($\bar{r} = 0.05454\%$, $s_r = 1.1392\%$, $m = 245$), let adjacent days have correlation 0.04 and other pairs none. Find the factor by which the adjacent correlations multiply Section 4's variance, and the year's standard deviation (two decimals).

6. On the TAIEX price index (no dividends), U28 dates, R_{t+1} had mean 0.2251% and standard deviation 1.8889% after the 306 days with $|R_t| \geq 2\%$. After the other 4,075 it had 0.0496% and 1.0570% (divisor each group's days). Weighting over the 4,381 pairs, find the mean (four decimals), the three parts of (16.6) (three decimals, %²) and the share between (in %, one decimal). What does Example 4's 0.040 miss?

7. With Example 7's inputs, let Semiconductor return −3%. Find FinancialInsurance's conditional mean (two decimals) and standard deviation; from the rounded mean, the standardized value of −2% (four decimals); and $\Pr(Y \leq -2\% \mid X = -3\%)$.

8. On 5 August 2024 the pair returned −9.48% and −6.93%. With Example 7's means and LM12's $l_{11} = 1.5243$, $l_{21} = 0.6998$, $l_{22} = 0.9960$ (% a day), compute z_1, z_2 (three decimals) and d^2. Then find the model's probability of a larger d^2 and the expected such days in 4,382.

9. Prove (a) $|\rho_{XY}| \leq 1$ from $\mathrm{Var}(X/\sigma_X \pm Y/\sigma_Y) \geq 0$, and say when $\rho_{XY} = 1$; (b) $\mathrm{Cov}(M\mathbf{X} + \mathbf{b}) = M\Sigma M^\top$, and that $\mathrm{Corr}(\mathbf{X})$ is positive semidefinite.

10. Evaluate (16.10) at $\rho = 0.5749$ with $\mathrm{Var}(Z \mid |Z| \geq 2) = 5.7464$ and with $\mathrm{Var}(Z \mid Z \leq -2) = 0.1143$ (three decimals), and compare Exhibit 8's data. Which event raises the benchmark above ρ, and why?

11. A normal pair has $\rho = 0.5749$. On the days when both variables fall 2 standard deviations or more below their means, their correlation is *closest* to:

A. 0.575

B. 0.156

C. 0.860

12. *(Python)* Rebuild Exhibit 2 with the guard, and again with exact returns, `Fraction` of the two-decimal closes, compared with `Fraction(-1, 100)` and `Fraction(1, 100)`. Compute the pair's d_t^2 with `np.linalg.cholesky` and `solve_triangular`, and compare the count of $d_t^2 > 9$ with `T *`

`stats.chi2.sf(9, 2)`. Explain why `1 - stats.chi2.cdf(89.30, 5)` and `stats.chi2.sf(89.30, 5)` differ.

SOLUTIONS

1. **B** is correct: $347/859 = 0.404$. A is the joint probability $347/4{,}382 = 0.079$. C conditions the wrong way, $347/538 = 0.645$.

2. **A** is correct. Here $\mathbb{E}X = \int_0^1 x(x + \frac{1}{2})\,dx = \frac{7}{12}$ and $\mathbb{E}[XY] = \int_0^1\int_0^1 xy(x+y)\,dy\,dx = \frac{1}{3}$, so $\mathrm{Cov}(X,Y) = \frac{1}{3} - \frac{49}{144} = -\frac{1}{144} = -0.0069$. With $\mathrm{Var}\,X = \frac{11}{144}$ the correlation is $-\frac{1}{11} = -0.0909$. B omits $\mu_X\mu_Y$; C assumes independence, which Example 2 excluded.

3. **C** is correct. By symmetry, $\Pr(Y \le y) = \frac{1}{2}\Phi(y) + \frac{1}{2}\Pr(-X \le y) = \Phi(y)$. By (16.1), $\mathrm{Cov}(X,Y) = \mathbb{E}[S]\,\mathbb{E}[X^2] = 0$, yet $|Y| = |X|$. A needs joint normality. B fails: $X + Y = 0$ with probability 1/2, while for a normal pair with $\rho = 0$, $X + Y \sim \mathcal{N}(0, 2)$ gives each single value probability 0.

4. By (16.3), $\mathrm{Var}(R_1 - R_2) = 0.04 + 0.0225 - 2(0.5)(0.20)(0.15) = 0.0325$, a volatility of 18.03%. Next, $\mathrm{Cov}(R_1, R_p) = 0.6(0.04) + 0.4(0.015) = 0.0300$, and $\sigma_p^2 = 0.36(0.04) + 2(0.24)(0.015) + 0.16(0.0225) = 0.0252$, so $\sigma_p = 15.87\%$. The correlation is $0.0300/(0.20 \times 0.1587) = 0.945$. The value 5% is $|\sigma_1 - \sigma_2|$, right only for $\rho = 1$; 25% is $\sqrt{\sigma_1^2 + \sigma_2^2}$, right only for $\rho = 0$.

5. By (16.3), the variance is ms_r^2 plus twice the $m - 1$ adjacent covariances $0.04s_r^2$. The factor is $1 + 2(0.04)(244/245) = 1.0797$, and the standard deviation is $17.83\% \times \sqrt{1.0797} = 18.53\%$. Wrong routes: 245 × 1.1392% = 279.1% adds standard deviations, and 17.83% × 1.0797 = 19.25% applies a variance factor to a standard deviation. This is a model; LM19 examines the data.

6. By (16.5), $(306 \times 0.2251 + 4{,}075 \times 0.0496)/4{,}381 = 0.0619\%$. By (16.6), within 1.288 plus between 0.002 gives the total 1.290 (%²), a share between of 0.2% (0.155%). The between part is tiny (0.002 of 1.290), but the spreads differ by a factor of 1.8889/1.0570 = 1.79, which a correlation of returns cannot see. The equal-weight mean, (0.2251 + 0.0496)/2 = 0.13735%, ignores that big days are rare.

7. By (16.8), the mean is $0.0494 + 0.4591 \times (-3 - 0.0947) = -1.37\%$, and the standard deviation stays 0.996%. Then $(-2 + 1.37)/0.996 = -0.6325$ and $\Phi(-0.6325) = 0.264$. The unconditional standard deviation would give $\Phi(-0.63/1.2173) = 0.302$, and the slope ρ in place of β a mean of −1.73%.

8. Forward substitution gives

$$\begin{aligned} z_1 &= (-9.48 - 0.0947)/1.5243 = -6.281, \\ z_2 &= (-6.93 - 0.0494 - 0.6998 z_1)/0.9960 \\ &= -2.594, \end{aligned}$$

   so $d^2 = 46.18$; unrounded inputs give 46.22. By (16.9), $\Pr(d^2 > 46.18) = e^{-23.09} = 9.4 \times 10^{-11}$, about 4×10^{-7} expected days in 4,382.

9. (a) By (16.3), $\mathrm{Var}(X/\sigma_X \pm Y/\sigma_Y) = 2 \pm 2\rho_{XY} \ge 0$, so $-1 \le \rho_{XY} \le 1$. If $\rho_{XY} = 1$, then $X/\sigma_X - Y/\sigma_Y$ has variance 0, so it is constant with probability one (Section 3's Deep Dive): $Y = \mu_Y +$

$(\sigma_Y/\sigma_X)(X-\mu_X)$. (b) The deviation $M(\mathbf{X}-\boldsymbol{\mu})$ gives $M\Sigma M^{\top}$ by linearity. With $M=\operatorname{diag}(\sigma_i)^{-1}$ this is $\operatorname{Corr}(\mathbf{X})$, positive semidefinite by (16.4).

10. By (16.10),

$$\begin{aligned}&\frac{0.5749}{\sqrt{0.5749^2+(1-0.5749^2)/5.7464}}\\&\quad=0.860,\end{aligned}$$

    and with 0.1143 the result is 0.231. The two-sided event widens the spread of X (variance 5.7464 against 1) and raises the correlation; the one-sided tail narrows it (0.1143) and lowers it. The data's 0.788 lies below 0.860, while their 0.488 lies far above 0.231.

11. **B** is correct, the benchmark of Exhibit 9 at $k=2$. A ignores the conditioning. C is the benchmark for moves of one variable either way (Problem 10).

12. Both methods give Exhibit 2, since no return is exactly ±1%. The pair has 177 days with $d_t^2>9$ against the model's 48.68. The difference `1 - cdf` returns 0.0 and `sf` returns 9.4e-18, as Section 8's pitfall explains.

GLOSSARY

Term	中文	Meaning
Joint distribution	聯合分布	The law of several random variables together: a joint pmf, density or distribution function.
Marginal distribution	邊際分布	The law of one variable, obtained from the joint law by summing or integrating the others out.
Conditional distribution	條件分布	The law of Y given $X = x$: the joint pmf or density divided by the marginal of X.
Independent	獨立	Random variables whose joint distribution function factorizes, (16.1).
Covariance	共變異數	$\mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$, (16.2).
Correlation	相關係數	$\sigma_{XY}/(\sigma_X \sigma_Y)$, between −1 and 1.
Uncorrelated	互不相關	With covariance zero; implied by independence, not conversely.
Covariance matrix	共變異數矩陣	$\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top]$, positive semidefinite, (16.4).
Conditional expectation	條件期望值	$\mathbb{E}[Y \mid X = x]$, the mean of the conditional law; $\mathbb{E}[Y \mid X]$ is that function of X.
Tower property	雙重期望值定理	$\mathbb{E}[\mathbb{E}[Y \mid X]] = \mathbb{E}Y$, (16.5).
Law of total variance	全變異數法則	Variance = mean within-group variance + variance of group means, (16.6).
Multivariate normal distribution	多變量常態分布	A random vector all of whose combinations are normal, $\mathcal{N}_N(\boldsymbol{\mu}, \Sigma)$, (16.7).
Colouring, whitening	著色、白化	$\boldsymbol{\mu} + L\mathbf{Z}$ and $L^{-1}(\mathbf{X} - \boldsymbol{\mu})$, with $\Sigma = LL^\top$.
Mahalanobis distance	Mahalanobis 距離	d, with $d^2 = (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})$.
Chi-square distribution	卡方分布	The law χ^2_N of a sum of N independent squared standard normals: mean N, variance $2N$.
Degrees of freedom	自由度	The number N of squared standard normals in χ^2_N.
Conditional correlation	條件相關係數	The correlation given an event, (16.10) for a normal pair.

Term	中文	Meaning
Exceedance correlation	超越相關係數	The correlation on the days when both variables lie beyond k standard deviations on the same side.

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